

Benchmarking computational methods for detecting spatial domains and domain-specific spatially variable genes from spatial transcriptomics data

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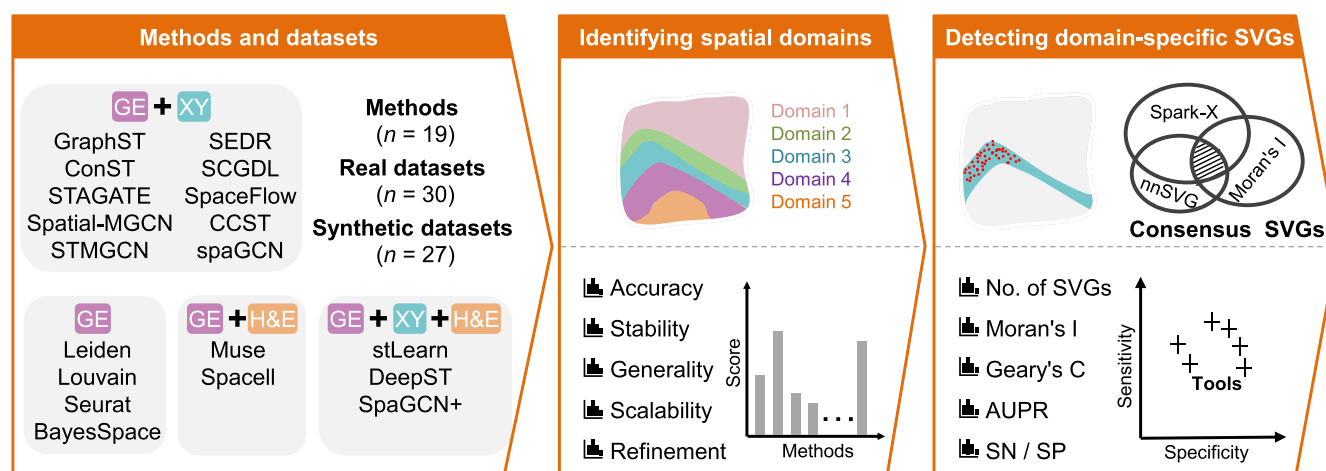
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Abstract

Advances in spatially resolved transcriptomics (SRT) have led to the emergence of numerous computational methods for identifying spatial domains and spatially variable genes (SVGs); however, a comprehensive assessment of existing methods is lacking. We comprehensively benchmarked 19 methods for detecting spatial domains and domain-specific SVGs from SRT data, using 30 real-world datasets covering six SRT technologies and 27 synthetic datasets. We first evaluated the performance of these methods on spatial domain identification in terms of accuracy, stability, generalizability, and scalability. Results reveal that there is no single method that works best for all datasets, and the optimal method depends on the data, especially the SRT platform. Further, we proposed a quantitative strategy to evaluate domain-specific SVG recognition results and assessed the impact of spatial domains on SVG detection. We found that SVG detection based on spatial domains identified by different GNN methods have high accuracy but low concordance. Generally, the more accurate the recognized spatial domains, the higher the number and accuracy of domain-specific SVGs detected. Moreover, integrating spatial clustering results from different methods can lead to more robust and better clustering and SVG results. Practical guidelines were provided for choosing appropriate methods for spatial domain and domain-specific SVG identification.

Graphical abstract



Introduction

Advances in spatially resolved transcriptomics (SRT) have made it possible to obtain spatial expression data and positional information of genes, which is crucial for the in-depth understanding of organ function and tissue microenvironment

[1]. SRT technologies can be divided into two main categories: imaging-based technologies [e.g. SeqFish [2], MERFISH [3], and STARmap [4]] and next-generation sequencing (NGS)-based technologies [e.g. ST [5], Slide-seqV2 [6], Stereo-seq [7], and 10X Visium [8, 9]. Imaging-based technologies can

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provide subcellular resolution data, but are often limited by low sequencing throughput and mRNA detection sensitivity, as well as high cost and time consumption [10]. In contrast, NGS-based technologies utilize spatial barcodes to capture RNA molecules at specific locations (known as spots) followed by sequencing, which typically have relatively low resolution but are favored for their high throughput, cost-effectiveness, and ability to provide whole genome gene expression data [10]. In particular, 10X Visium and Stereo-seq have achieved remarkable commercial success due to their cost-effectiveness and wide range of applications [11].

Advances in SRT have generated vast and complex data, leading to the emergence of numerous computational methods for analyzing SRT data [12]. SRT data allow for solving a range of distinct questions, among which spatial domain identification (commonly referred to as spatial clustering) is a prerequisite for investigating spatial transcriptomics. Spatial domain identification delineates spatially coherent regions in a tissue based on SRT data, which have been indicated crucial in identifying domain-specific marker genes [13], deciphering cell types [14], analyzing cell-to-cell communication [15], and profiling the emergence and progression of disease [16, 17]. In addition, spatial variable gene (SVG) identification is also a fundamental step in the downstream analysis of SRT data, which refers to the identification of genes that exhibit spatial patterns across the entire tissue section or within predefined spatial domains. SVGs that exhibit global, continuous variations across the entire spatial domain refer to as global SVGs, while SVGs characterized by variations confined to predefined specific spatial domains refer to as spatial domain-specific SVGs. SVG detection is essential for capturing genes carrying biological signals, revealing new therapeutic targets or prognostic markers [18, 19]. Particularly, spatial domain-specific SVGs can serve as landmarks for restoring spatial locations of cells in single-cell RNA-seq [20, 21]. As SRT technology continues to advance, SRT analysis methods are constantly emerging and evolving towards two remarkable trends [22, 23]. One trend is to shift from relying on single gene expression information to integrating multimodal data, such as Hematoxylin and Eosin (H&E) image information and spatial locations. Another trend is the transition from traditional machine learning methods to deep learning methods, especially graph neural networks (GNNs), which are highly valued for their ability to automatically extract features and handle complex data relationships.

Although numerous methods are available for detecting spatial domains and SVGs [23, 24], studies corresponding to different methods often use limited datasets and/or evaluation metrics for performance evaluation, which poses great challenges in choosing appropriate methods for analyzing SRT data generated by certain laboratories. Therefore, it is necessary to benchmark these emerging methods with more comprehensive datasets and performance metrics to provide practical guidance on data analysis for the biological community. In terms of spatial clustering evaluation, Cheng *et al.* [25] first evaluated five early methods using seven semi synthetic datasets and five real datasets. Subsequently, Liu *et al.* [22] evaluated the clustering performance of eight representative GNN methods. These earlier evaluation studies only considered limited methods and datasets, and did not evaluate the impact of clustering on downstream SVG recognition. For SVG identification evaluation, recent benchmark studies [10, 26, 27] compared 6–14 traditional statistical methods to iden-

tify SVGs. However, these studies only focused on methods detecting global SVGs rather than spatial domain-specific SVGs. Moreover, they evaluated the performance of SVG identification on real data by using metrics such as Moran's I and Geary's C to measure the spatial autocorrelation of SVGs [28, 29], or by conducting gene ontology (GO) analysis to explore their biological significance [30]. Consequently, they are unable to assess the accuracy of each SVG or the quantity of SVGs, and thus cannot quantitatively measure the validity of the identified SVGs. Lately, Yuan *et al.* [31] provided a detailed benchmark of spatial clustering methods and further evaluated the ability of these methods to identify domain-specific marker genes. However, Yuan *et al.* only utilized Seurat [21] for marker gene identification and did not consider SVG recognition methods specifically targeting spatial transcriptomes. Moreover, Yuan *et al.* also only relied on Moran's I and Geary's C to evaluate marker genes. Overall, existing benchmark studies mainly consider clustering or SVG recognition as a single task without delving into the impact of clustering on downstream SVG recognition and lack quantitative indicators to evaluate the effectiveness of SVG recognition.

In this study, we benchmarked 19 computational methods, including 14 newly emerging GNN methods, for detecting spatial domains and domain-specific SVGs from spatial transcriptomics data, using 30 real-world datasets covering six SRT technologies and 27 synthetic datasets generated by two simulation tools. We first comprehensively evaluated the performance of 19 methods on identification of spatial domains, in terms of accuracy, stability, generalizability, and scalability. Our results reveal that there is no single method that works best for all datasets, and the optimal method depends on the data, especially the SRT platform that generates the data. Moreover, we extended the framework of the GNN methods by considering factors such as input genes, clustering algorithms, and refinement options, and found that the default settings of many GNN methods are not always optimal and there is still room for improvement. Further, we proposed a quantitative strategy to evaluate SVG recognition results and assessed the impact of spatial domains identified by different methods on domain-specific SVG detection. We found that detection of domain-specific SVGs based on spatial domains identified by different GNN methods have high accuracy but low concordance, and the parameters of the SVG detection method have great impact on the SVG detection results. Particularly, although SVG results depend on specific data, overall, the more accurate the recognized spatial domains, the higher the number and accuracy of domain-specific SVGs detected. Moreover, integrating spatial clustering results from different GNN methods can lead to more robust and better clustering and SVG results. Taken together, our work provides useful information for choosing and developing appropriate methods to identify spatial domains and domain-specific SVGs for data from different SRT platforms.

Materials and methods

Datasets

A total of 30 real-world datasets were used in this study (Supplementary Table 1), involving data from six SRT technology platforms, including ST, 10X Visium, Slide-seqV2, STARmap, SeqFISH, and Stereo-seq. The main difference between the data of different SRT technology platforms is res-

olution. The resolutions of ST, 10X Visium, Slide-seqV2, STARmap, SeqFISH, and Stereo-seq are approximately 100 μm , 55 μm , 10 μm , subcellular, subcellular, and subcellular levels, respectively [32]. To reduce experimental bias caused by the same type of data, each type of SRT used in this study includes at least two slices from different tissues, rather than multiple replicate slices from the same tissue.

In addition, we generated 27 simulated datasets using two simulation tools, scDesign3 [33] and SRTsim [34] (Supplementary Table S2). Following a previous study [26], we used the scDesign3 tool to generate simulated datasets for human dorsolateral pre-frontal cortex (DLPFC) data with a ratio of SVGs to non-SVGs being 1:9 (denoted as DLPFC_scDesign). Specifically, we selected 200 SVGs from 2000 highly variable genes (HVGs), and the remaining genes were averaged to serve as non-SVGs. The simulated data retained the spatial locations and labels of the original data. Moreover, based on the DLPFC slice #151 673, we also generated additional simulated datasets (DLPFC_scDesign_2) with the number of reference SVGs ranging from 100 to 900 (in increments of 200), while keeping the total number of HVGs at 2000. When evaluating the SVG recognition performance of different GNN methods, we found that all GNN methods obtained very few SVGs for the first 8 slices of the DLPFC data (with an average of less than 20 SVGs). Therefore, we used all slices for evaluating spatial clustering while only used the last four slices for evaluating SVG detection. In addition to the scDesign3 simulation data, we also collected a breast cancer dataset generated by SRTsim tool (BC_SRTsim) [10]. BC_SRTsim datasets have different gene expression differences, allowing us to assess the impact of variation of gene expression on SVG identification. Besides, we also employed the SRTsim tool to generate six datasets based on the DLPFC #151 673 slice with log fold change (logFC) values ranging from 0.5 to 3 (DLPFC_SRTsim).

Methods for spatial domain identification

A total of 19 spatial clustering methods were benchmarked in this study (Supplementary Table S3), which are categorized into four groups based on their inputs.

(i) Group GE: unimodal methods that rely only on gene expression, including Leiden, Louvain, and Seurat.

Louvain [35] and Leiden [36] are two community detection algorithms, and Leiden is an improvement of Louvain. Both methods are often applied to the task of cell clustering in scRNA-seq analysis.

Seurat [21] is a comprehensive R package dedicated to scRNA data. It provides a complete set of analysis tools covering key steps from data preprocessing, clustering, differential expression analysis, and visualization. Seurat uses the Louvain algorithm for clustering by default.

(ii) Group GE + XY: methods considering gene expression and spatial locations, including BayesSpace, ConST, STAGATE, SEDR, SpaceFlow, Spatial-MGCN, STMGCN, SCGDL, CCST, and GraphST.

BayesSpace [37] utilizes spatial neighborhood structure to improve the resolution of the data and to perform clustering analysis. BayesSpace is based on the Bayesian statistical framework and performs spatial clustering through Markov random fields and Gibbs sampling.

ConST [38] is an interpretable contrast learning framework. It first extracts morphological features through a

mask autoencoder (MAE), captures spatial relationships using KNN, and generates graph embeddings using a variational graph autoencoder (VAE). The framework obtains potential representations through triple contrast learning (local-local, local-global, and local-contextual).

STAGATE [39] transforms spatial location information into a spatial neighbor network and combines this network with gene expression data. A low-dimensional representation of the spatial and expression information is extracted by an adaptive graph attention autoencoder. This representation can be used for a variety of downstream tasks, including spatial domain identification, data noise reduction, and 3D spatial domain extraction.

SEDR [40] learns low-dimensional feature vectors from gene expression data via a deep self-encoder and integrates spatial neighborhood relations using a variational graph self-encoder. The final joint embedding of gene expression and spatial information to improve clustering, visualization, trajectory inference, and batch effect correction.

SpaceFlow [41] first constructs a Spatial Expression Graph (SEG) and then encodes the SEG using Graph Convolutional Network (GCN) to generate embeddings that capture local expression patterns. To ensure spatial consistency, the embeddings are spatially regularized. Finally the low-dimensional embeddings are output for visualization, domain segmentation (clustering) and output pseudo spatio-temporal graphs.

Spatial-MGCN [42] is a spatial clustering algorithm based on multi-view GCNs. The algorithm uses gene expression profiles and spatial information to construct two neighbor graphs respectively, followed by a GCN encoder to extract gene expression information, spatial information. Further, Spatial-MGCN utilizes the attention mechanism to adaptively fuse these features. Subsequently, the algorithm employs a zero-inflated negative binomial decoder to reconstruct the feature matrix and capture the global information of the original spatial expression profile. Finally, spatial regularization constraints are incorporated into the representation learning process to preserve spatial neighbor information.

STMGCN [43] constructs neighbor views with both Euclidean and cosine similarity based on spatial coordinates. Then, STMGCN learns view-specific embeddings by combining gene expression with each neighboring graph via a GCN. Finally, to capture the importance of different graphs, it further introduces an attention mechanism to adaptively fuse to get the final embedding representation.

SCGDL [42] uses RGGCNN (Residual Gated Graph Convolutional Neural Network) as an encoder for the depth graph model to achieve higher-level mixing of gene expression profiles and spatial locations. Then, a Bayesian Gaussian Mixture Model is applied to process potential embeddings to generate spatial domains.

CCST [13] is similar to SCGDL in that both are based on the Deep Graph Infomax (DGI) [44] network. This method uses the DGI module to encode spatial structure and gene expression information. After dimensionality reduction by Principal Component Analysis, the K-means algorithm is used for node clustering to identify cell groups or subpopulations.

GraphST [45] is a spatial clustering framework using comparative learning techniques for self-supervised graphs. It combines GNNs with self-supervised contrast learning to learn a discriminative representation of Spot by minimizing the embedding distance between spatially adjacent points.

SpaGCN [46] uses gene expression, spatial location, and histological data (optional) to construct a weighted undirected graph of spots and performs cluster analysis on the constructed graph. SpaGCN originally performs end-to-end clustering. But in order to make a fair comparison with other methods, the final results it learns are saved as low-dimensional vectors for uniform comparison.

(iii) Group GE + H&E: methods considering H&E and gene expression, including MUSE and SpaCell.

MUSE [47] learns H&E and gene expression information by using two auto-encoders (AE). The model is trained by utilizing self-supervised loss, triple loss function.

SpaCell [48] integrates H&E images and spatial gene expression data to classify cell types and disease stages by training neural network models. For H&E images, SpaCell extracted initial image features using a pre-trained CNN (e.g. ResNet50). Subsequently, AE was used to find the latent space of H&E features and gene expression data and combine them. However, the method did not use the spatial information of the case, and the information of the two features is only simply concatenated, and has not yet been sufficiently interactively fused.

(iv) Group GE + XY + H&E: multimodal methods that consider gene expression, spatial locations, and H&E image, including SpaGCN+, stLearn, and DeepST.

SpaGCN+ [46], a variant of SpaGCN that uses gene expression, spatial location, and H&E image information to detect spatial domains.

stLearn [49] developed a spatial morphological gene expression (SME) standardization method to recalculate gene expression values by averaging neighboring points. It homogenizes gene expression data using information about the spatial location of neighbors and morphological distances.

DeepST [50] integrates H&E, gene expression, and spatial location data using a Domain Adversarial Neural Network (DAN). Potential representations of augmented spatial transcriptome data are jointly generated using an autoencoder and a denoising autoencoder.

Unless otherwise specified, we used the default parameter settings and preprocessing of all methods. The setting of the number of spatial domains varies depending on different datasets. For algorithms that can directly input the number of spatial domains (e.g. GraphST and STMGCN), we set the number of clusters directly based on annotated categories of the data. For algorithms that accept a clustering resolution parameter instead, we search for the resolution that most closely aligns with the desired number of spatial domains. The specific parameter settings for each benchmarking method are summarized in [Supplementary Table S4](#).

Next, to provide deeper insights and stimulate further innovations and improvements in the field, we here attempt to summarize the historical development and interrelationships of above methods. SpaGCN, an early spatial clustering method, leveraged GCNs to aggregate information from neighboring spots. However, its fixed adjacency relationships limited adaptability to the similarities between spots and failed to fully exploit features from histological images. Later, STAGATE significantly improves on SpaGCN, which adopts an attention mechanism to adaptively learn the similarity of neighboring spots. GraphST further develops on STAGATE by adopting a self-supervised contrastive learning mechanism, which enhances the ability to capture local contextual infor-

mation and improves the clustering performance. In terms of utilizing H&E images, SpaGCN constructs a weighted graph by integrating the RGB features of H&E images with spatial location information to fully leverage H&E information. In contrast, DeepST and stLearn extract H&E features using ResNet-50 and compute the similarity of the H&E feature matrix as a constraint to augment or normalize the gene expression matrix. The latest methods, such as stGCL [51] and TransformerST [52], employ Visual Transformers (ViT) and their variants (H-ViT) to more accurately extract rich texture features from H&E images, further improving the extraction and integration of H&E information. In terms of framework design, many methods have attempted to replace GCN with graph autoencoder to improve the GCN network in SpaGCN. For example, conST uses a Masked Autoencoder (MAE), while SEDR employs a Deep Autoencoder and a VAE for encoding. Additionally, methods such as SpaceFlow, CCST, and SCGDL incorporate contrastive learning techniques, utilizing DGI networks to maximize the mutual information of nodes in the graph, thereby effectively learning low-dimensional representations of the nodes. ConST adopts triple contrast learning (local-local, local-global, and local-contextual) to further enhance representation learning, primarily focusing on spatial clustering. Furthermore, GraphST combines self-supervised contrastive learning with flexible module design, enabling it to handle various analytical tasks, including clustering, batch integration, and deconvolution.

Assess the impact of the choice of clustering method, refinement, and hyperparameters on spatial domain identification

We evaluate 12 GNN methods capable of generating embedded representations. Instead of using the default clustering algorithm in each GNN method, we applied four different clustering algorithms, including mclust (model-based clustering), K-means, Louvain, and Leiden. Moreover, we further conducted a refinement step for all GNN methods, which is a common post-processing strategy of spatial clustering used in many GNN methods such as GraphST, SpaGCN, and DeepST. The refinement step refines the spatial clustering results by re-assigning the labels of spots to the most common labels among adjacent spots, which helps to make the domain boundaries smoother. In this study, we utilized the *refine_label* function from GraphST to refine results from each GNN method to ensure fairness. Finally, a total of eight configurations were generated, covering four clustering algorithms in default or refined mode.

Moreover, to further validate the impact of hyperparameters in GNN-based methods on model performance, we selected three common hyperparameters—number of epochs, low-dimensional embedding dimensions, and learning rate—to conduct experiments on the DLPFC #151 673 dataset. Specifically, we selected 11 representative GNN-based methods (Stlearn was excluded because it does not work with GNN models). The hyperparameters were configured as follows: the number of epochs was set to [200, 400, 600, 800, 1000], the range of input dimensions was [200, 300, 400, 600, 800, 1000, 3000], and the learning rate was set to [0.0001, 0.001, 0.01, 0.1]. During the evaluation of one hyperparameter, the other two parameters were held at their default values ([Supplementary Table S4](#)).

Identification of spatial domain-specific SVGs

Spatial domain-specific SVGs were detected based on spatial domains identified from different spatial clustering methods. We followed the SVG identification function provided in SpaGCN [46] to identify SVGs that are specific to a spatial domain. Unless otherwise specified, we used the default parameter settings provided in SpaGCN (P -value < 0.05 , in out group ratio > 1 , in group fraction > 0.8 , and $\log FC > 1.5$). Four filtering criteria were used in SpaGCN. First, the genes must pass the test of statistical significance, i.e. P -value < 0.05 . Second, the genes must be expressed within the target domain significantly higher than outside the domain. Third, the proportion of gene expression in cells of the target domain needs to know needs to reach 80%. Fourth, the expression level of the gene in the target domain is at least 1.5 times higher than that of cells in other domains.

Constructing consensus SVGs for real SRT datasets

Since real SVGs are unknown in most real data, we first proposed a global-SVG strategy to construct consensus global SVGs as the reference for evaluating domain-specific SVGs on real datasets. Consensus SVGs refer to SVGs recognized by multiple tools, which are considered more reliable. Based on previous global SVG benchmarking studies [10, 26] and our practical experience, we selected three high-performance SVG detection methods, including SPARK-X [53], nnSVG [54], and Moran's I [55]. In addition to these three tools, SOMDE [56] and SpatialDE2 [57] were also ranked top in previous benchmarking studies [10, 26]. However, our initial evaluation revealed that SOMDE identified fewer SVGs in some datasets, leading to a limited number of consensus SVGs among different tools (Supplementary Fig. 1). SpatialDE2 does not perform significance testing, making it difficult to determine the number of SVGs. It is reassuring that we have also replaced SPARK-X with SpatialDE2 in the SVG toolset to compare the obtained consensus SVGs. We followed the previous benchmarking study [10] and selected the top 2000 genes based on the spatial variance score (FSV) as SVGs. The results showed that although the number of SVGs obtained by SpatialDE2 and SPARK-X on the same dataset differs significantly (Supplementary Figs S2–S3), the consensus SVG sets obtained by these two tools through intersection with other two tools are very close (Supplementary Fig. S2b). This result again confirms the necessity of using different tools to jointly obtain the consensus SVG set. Finally, we considered the intersection of the recognition results from the three methods (SPARK-X, Moran's I, and nnSVG) as the consensus SVG set. It should be noted that the number of SVGs recognized by different SVG recognition methods may vary greatly on some data, resulting in very few consensus SVGs, even less than the number of SVGs recognized by the evaluated GNN methods. In these cases, we used a looser strategy to construct the consensus SVG set. For example, for the ST_Hip_2 and Stereo_AB data, the number of consensus SVGs is low (only 85 and 101, respectively), even less than the number of SVGs identified by some benchmarked GNN methods (e.g. Spatial-MGCN = 141, STAGATE = 102 for the ST_Hip_2 dataset; Spatial-MGCN = 336, STAGATE = 145 for the Stereo_AB dataset). For these two datasets, we first took the union of SVGs identified by nnSVG and Moran's I, and then intersected the union with SVGs identified by SPARK-X to obtain consensus SVGs.

By flexibly combining different tools as described above, the false positives (FPs) in the consensus SVG set can be minimized, resulting in a conservative SVG set. However, it is inevitable that some real SVGs that are not recognized by all methods may also be missed, making the consensus set unable to always cover all real SVGs. Indeed, by careful inspection, we found that some domain-specific SVGs identified by GNN methods that are not present in the consensus SVG set may actually exhibit spatial patterns. For example, in the 10X_151_673 dataset, two genes, MT-ND2 and FTH1, detected based on the spatial domains identified from stLearn and STAGATE exhibit distinct spatial patterns and have high Moran's I score, whereas they were not in the consensus SVG list (Supplementary Fig. S4a). Moreover, through the analysis of simulation data of DLPFC with 200 known benchmark SVGs, we also found that consensus SVGs cannot fully cover the benchmark SVGs, accounting for less than half of the benchmark SVGs (Supplementary Fig. S4b). Therefore, to avoid bias in evaluation results caused by the incompleteness of the consensus SVG set, we also consider the SVGs identified by GNN methods that are not in the consensus set but with high Moran's I score as true positives (TPs). Specifically, for each dataset, we used the average Moran's I score of true SVGs (i.e. those in the consensus SVG set) identified by all GNN methods as the reference Moran's I score. Any SVGs identified by a GNN method not in the initial consensus SVG set but with a Moran's I score greater than the reference score were reclassified as TPs. The reference Moran's I score for each data is provided in Supplementary Table S5. This refinement strategy ensures that the final constructed consensus set contains as many genes with spatial expression pattern as possible.

Metrics for assessing spatial domain identification

We used three external metrics considering true labels to assess the accuracy of spatial clustering, including Adjusted Rand Index (ARI), Normalized Mutual Information (NMI), and Adjusted Mutual Information (AMI). In addition, we used an internal metric, Silhouette Coefficient (SC), which does not require true labels to assess the separation of clusters.

ARI is based on the adjustment of the Rand Index (RI) (equation 1). $E(RI)$ is the expected value of RI and $\max(RI)$ is the maximum possible value of RI .

$$ARI = \frac{RI - E(RI)}{\max(RI) - E(RI)} \quad (1)$$

RI is defined as the similarity between two clusters (equation 2), where U represents the actual categorization and C represents the clustering result. a is the number of objects in the same class in U and in the same class in C . b is the number of objects in the same class in U but in different classes in C .

$$RI = \frac{a + b}{C_N^2} \quad (2)$$

NMI measures the amount of shared information assigned to the two clusters and is normalized to the interval $[0, 1]$ (equation 3); $I(X; Y)$ is the mutual information of X and Y , and $H(X)$ and $H(Y)$ are the entropy of X and Y , respectively.

$$NMI = \frac{2 \times I(X; Y)}{H(X) + H(Y)} \quad (3)$$

AMI, like NMI, is based on mutual information (equation 4). However, AMI gives a greater penalty for random segmentation, which better avoids the effect of cluster randomness on the evaluation results.

$$AMI = \frac{I(X; Y) - E(I(X; Y))}{\max(H(X), H(Y)) - E(I(X; Y))} \quad (4)$$

Here $I(X; Y)$ is the mutual information of X and Y ; $E(I(X; Y))$ is the expected value of the mutual information, and $\max(H(X), H(Y))$ is the value of the larger entropy in X and Y .

SC (equation 5) does not depend on annotation labels and evaluates the clustering results based on the closeness of each sample to the cluster to which it belongs and its separation from other clusters.

$$SC = \frac{1}{n} \sum_{i=1}^n \frac{b_i - a_i}{\max(a_i - b_i)} \quad (5)$$

Here a_i is the average distance between the i -th sample and its nearest in-cluster sample, and b_i is the average distance between the i -th sample and its nearest out-of-cluster sample. SC measures the difference between the similarity of a data point to other points in its own cluster and its dissimilarity to points in the nearest cluster.

Metrics for assessing spatial domain-specific SVG identification

We measured the accuracy of the model in recognizing SVG by calculating the area under the curve of the model's precision and recall at different thresholds. We then followed the idea in a previous study [17] to calculate the AUPR (Area Under the Precision-Recall Curve). The AUPR, ranging from 0 to 1, quantifies the trade-off between precision and recall. Higher values indicate superior classifier performance.

Sensitivity (SN), specificity (SP), and precision were also used to evaluate the SVG detection results (equation 6).

$$\begin{aligned} SN &= \frac{TP}{TP+FN} \\ SP &= \frac{TN}{TN+FP} \\ Precision &= \frac{TP}{TP+FP} \end{aligned} \quad (6)$$

Moran's I (equation 7) is a metric that quantifies the degree of spatial autocorrelation in spatial statistics and was used to assess the spatial autocorrelation of each SVG computed by each method. The values of Moran's I can range from -1 to 1 . Values close to 1 indicate clear spatial patterns, whereas values close to 0 indicate random spatial representations, and values close to -1 indicate checkerboard-like patterns.

$$I = \frac{N}{W} \frac{\sum_{i=1}^N \sum_{j=1}^N wij (x_i - \bar{x})(x_j - \bar{x})}{\sum_{i=1}^N (x_i - \bar{x})^2} \quad (7)$$

Here x_i and x_j are the gene expression values of cell i and cell j , $\bar{x}$ is the average gene expression value of the gene; N is the number of cells and W is the sum of wij . If cell j is in the nearest neighbor of cell i , then we assign $wij = 1$, otherwise we assign $wij = 0$.

Geary's C (equation 8), like Moran's I, is a metric for calculating the degree of spatial autocorrelation. Geary's C is between 0 and 2 .

$$C = \frac{N}{2W} \frac{\sum_{i=1}^N \sum_{j=1}^N wij (x_i - \bar{x})(x_j - \bar{x})}{\sum_{i=1}^N (x_i - \bar{x})^2} \quad (8)$$

To align Geary's C with Moran's I on the same scale, we transform it from the range $[0, 2]$ to the range $[-1, 1]$ using the formula $C^* = 1 - C$.

The Agreement metric [30] evaluates the consistency of SVGs identified by different methods for a given dataset (equation 9).

$$Agreement = \frac{1}{|S|} \sum_{gi \in S} \frac{n_{gi} - 1}{N - 1} \quad (9)$$

N is the number of SVG recognition methods, S denotes the total number of genes, consisting of the concatenated set of SVGs recognized by all methods, and n_{gi} is the number of methods recognizing gi genes. The metric ranges from 0 to 1 , where a value of 0 means that the sets of SVGs in all methods are completely disjoint from each other, and a value of 1 means that the sets of SVGs are the same in all methods.

Results

Benchmarking framework

We proposed an integrated evaluation framework for benchmarking 19 methods on identifying spatial domains and spatial domain-specific SVGs (Fig. 1). We have collected 30 real datasets covering six SRT technologies and 27 synthetic datasets simulated by two methods (Fig. 1a). The 19 methods are categorized into four groups based on data input: (i) single-modality methods relying solely on gene expression (referred to as GE methods); (ii) methods combining H&E images with gene expression (referred to as GE + H&E methods); (iii) methods based on gene expression and spatial location (referred to as GE + XY methods); (iv) multimodal methods that integrate gene expression, H&E images, and spatial location information (referred to as GE + XY + H&E methods) (Fig. 1b). We evaluated the performance of different methods in spatial domain recognition from four different aspects, including accuracy, stability, adaptability, and scalability (Fig. 1c). We also assessed the impact of clustering methods and refinement on spatial domain identification. Furthermore, we evaluated the impact of spatial domains identified by different methods on domain-specific SVG detection (Fig. 1d). To evaluate the performance of SVG detection on real datasets, we proposed a strategy to construct consensus SVGs using global SVG detection methods as the reference and corrected the consensus set by calculating the Moran's I score. We comprehensively evaluated SVGs detected based on spatial domains identified by different methods using different indicators such as sensitivity and specificity. Furthermore, we explored how spatial domain recognition results affect the number of detected SVGs.

Evaluating the accuracy and stability of methods in spatial domain identification using the human DLPFC dataset

The DLPFC dataset with 12 tissue slices has been used in various studies for evaluating spatial domain identification due to its high-quality annotation and clear spatial structure [58]. Here we first used the slice 151 673 of DLPFC dataset to evaluate the performance of spatial domain identification of the different methods, using four clustering metrics, ARI, NMI, AMI, and SC. Among these metrics, the trends of ARI, NMI, and AMI are consistent (Fig. 2a and b and Supplementary Fig. S5a and b), so we mainly focus on ARI

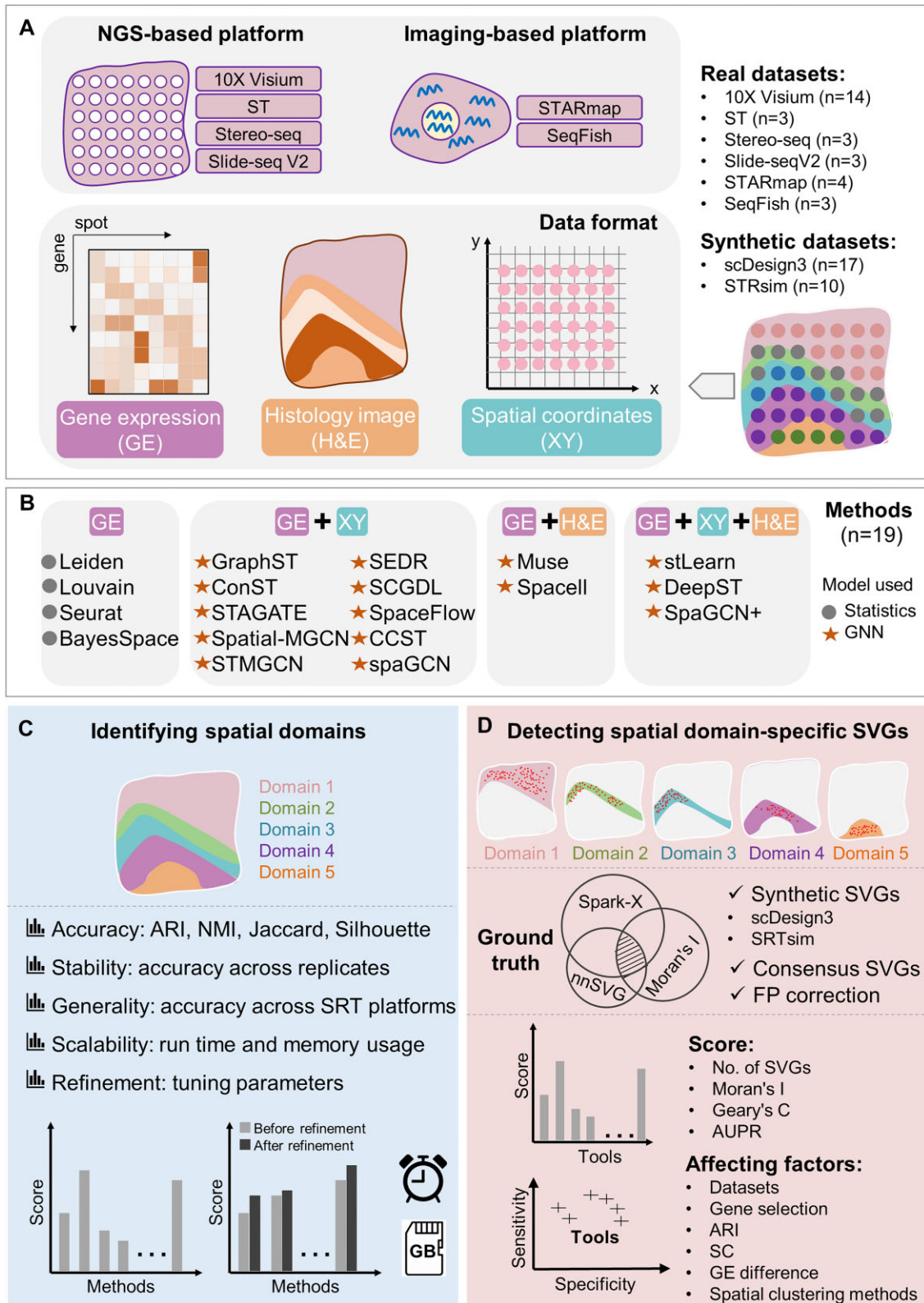


Figure 1. Benchmarking framework. **(A)** Summary of real and simulated datasets used in this study. **(B)** Nineteen methods are categorized into four groups based on data input. **(C)** Evaluation of the performance of different methods in spatial domain recognition. **(D)** Evaluation of the impact of spatial domains identified by different methods on spatial domain-specific SVG detection.

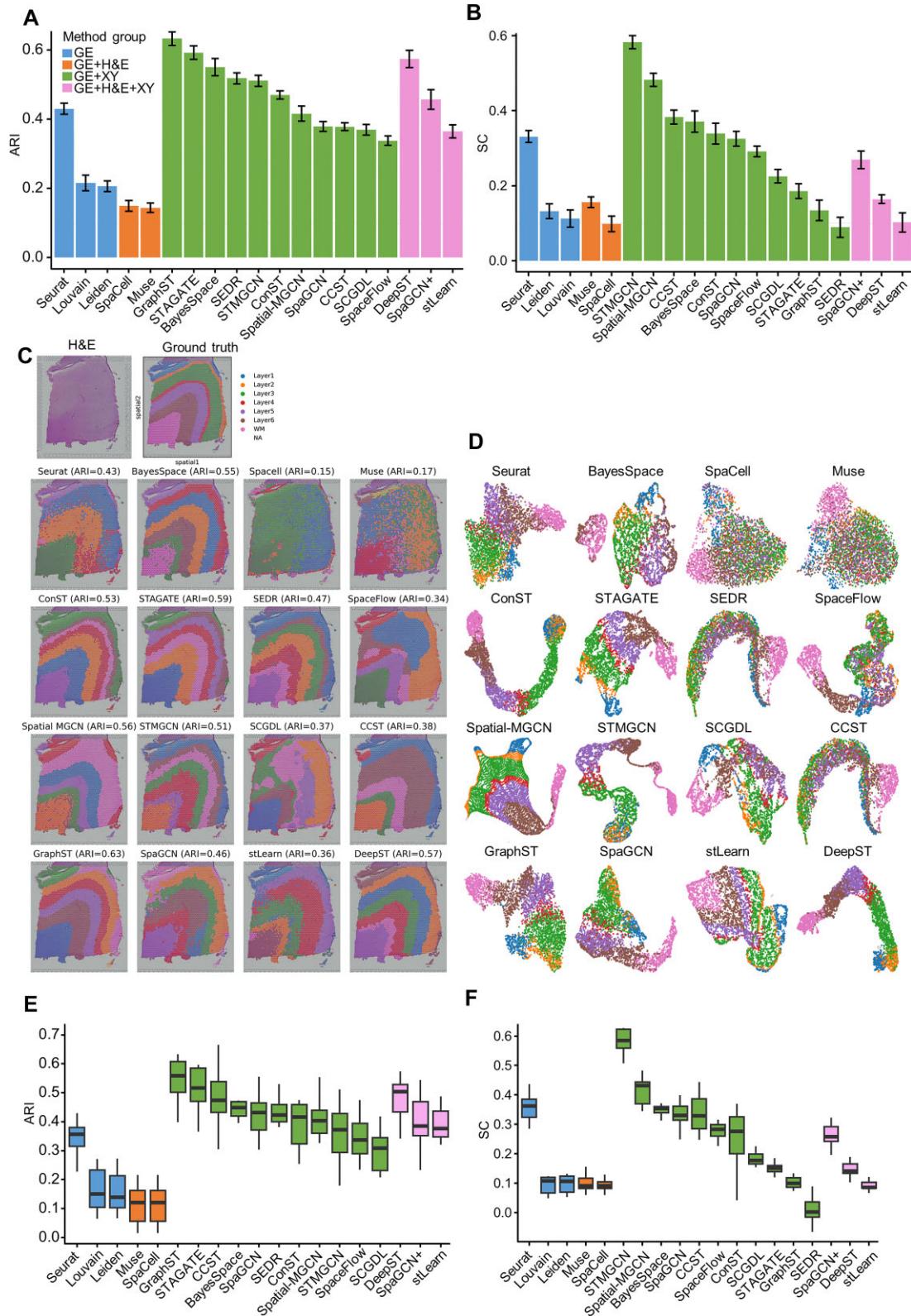


Figure 2. Evaluating the accuracy and stability of methods in spatial domain identification using the human DLPFC dataset. **(A)** ARI scores of different methods on the DLPFC#151 673 slice. The results of the same method group are depicted by the same color. **(B)** SC scores of different methods on the DLPFC#151 673 slice. **(C)** UMAP plots of 2D-embeddings obtained by different methods on the DLPFC#151 673 slice. **(D)** Visualization of each method detected spatial domains and the ground truth on the DLPFC#151 673 slice. **(E)** Summary of ARI scores of different methods on DLPFC 12 slices. **(F)** Summary of SC scores of different methods on DLPFC 12 slices.

and SC in subsequent analysis. GraphST outperforms other methods in terms of ARI (ARI = 0.633), indicating the highest similarity of predicted labels with the true labels. Meanwhile, STMGCN scores the highest on the SC metric (SC = 0.583), which means that its clustering results have better separation. Through UMAP (Uniform Manifold Approximation and Projection) visualization, it can be also clearly seen that spatial domains identified by GraphST have higher consistency with annotation (Fig. 2c), while results of STMGCN exhibit a more pronounced separation between clusters (Fig. 2d). Notably, SpaCell and MUSE, two methods that integrate gene expression and H&E information (group GE + H&E), performed the worst on this dataset. Of note, MUSE and SpaCell are early attempts to explore the use of H&E images, which showed promising results on simulated data or low-resolution data. However, these tools simply concatenate H&E and gene expression features for clustering, which may be ineffective for newly emerging, more complex ST data. For methods using only gene expression for clustering (group GE), Seurat provides superior performance compared to Leiden or Louvain. Moreover, methods of group GE are typically less effective than methods that include spatial location information (group GE + XY or GE + XY + H&E), highlighting the importance of spatial location information in clustering. In addition, by comparing the clustering results of the GE + XY and GE + XY + H&E groups, we found that methods considering H&E information did not yield overall better results than methods without considering H&E images. These results indicate that gene expression and spatial location information are crucial for spatial domain identification, while the impact of histology information is uncertain, which may depend on datasets and how image features are extracted and integrated.

Next, we assessed the robustness of different methods in spatial domain identification using 12 slices from the DLPFC dataset. Except for SpaGCN, the overall trend of different methods using all slices is the same as that using only slice 151 673 (Fig. 2e and f and [Supplementary Fig. S5c](#) and d). On slice 151 673, the version of SpaGCN with H&E images (SpaGCN+) performed better than SpaGCN without images (SpaGCN) (Fig. 2a). However, SpaGCN generally outperformed SpaGCN + considering all slices (median ARI: SpaGCN = 0.44, SpaGCN+ = 0.39). Specifically, out of the 12 slices, SpaGCN + outperformed SpaGCN on five slices, but underperformed on the remaining seven slices ([Supplementary Fig. S5e](#)). This finding suggests again that incorporating H&E images for SpaGCN may have limited contribution to the recognition spatial domain, and in some cases may even have a negative impact on clustering results.

Evaluating the generalizability of methods in spatial domain identification using datasets from different SRT technologies

Next, we utilized 30 datasets covering six SRT technologies to assess the adaptability of these methods for different SRT platforms. These datasets span a variety of SRT technologies, species, tissues, ensuring a broad and representative study ([Supplementary Table S1](#)). It is worth noting that due to the unavailability of H&E images for data from image-based SRT platforms, methods relying on H&E images (Muse, SpaCell, SpaGCN+, sLearn, and DeepST) were excluded from the evaluation in this section. All methods exhibit great fluctuations in clustering effects across different SRT datasets (Fig. 3a and

[Supplementary Fig. S6a](#)). Among the six SRT platforms, most methods performed better on 10X Visium and ST, while performing the worst on STARmap and Slide-seqV2. The inferior clustering for STARmap data may be attributed to the limited number of cells and genes ([Supplementary Fig. S6b](#)). In the case of Slide-seqV2 data, the poor clustering performance might be due to the relatively high proportion of zeros present in this type of dataset ([Supplementary Fig. S6c](#)). Spatial visualization of spatial domains recognized by different GNN methods for representative datasets from different platforms were shown in [Supplementary Figs S7–S10](#).

To more intuitively compare the clustering effects of different methods, we summarized the ranking of ARI scores for different methods (Fig. 3b). Overall, GraphST ranks first, closely followed by BayesSpace, SpaGCN, and STAGATE, all of which generally performed well on different datasets. SCGDL ranked the last, with general poor ARI scores across all datasets. Notably, the correlation between clustering results from different SRT platforms is relatively weak (Fig. 3c). However, within the same SRT platform, particularly for 10X Visium and SeqFish, a strong correlation was observed (Fig. 3c). Considering the differences in clustering performance across different SRT platforms, we further summarized the ranking of ARI scores for different methods within each SRT platform (Fig. 3d and [Supplementary Fig. S11](#)). For 10X Visium datasets, GraphST performed the best (mean ARI = 0.552), followed STAGATE (0.515) and CCST (0.481) (Fig. 3d). For STARmap data, ARI scores of the same method vary greatly across different datasets from the same SRT platform, with the score on dataset STARmap_MVC generally higher than the other three datasets. This also highlights the necessity of using diverse datasets for evaluation. Moreover, the performance of all methods on STARmap datasets is generally poor, with the best performing STAGATE having an average ARI score of only 0.307. For Slide-seqV2 data, the situation is even worse, with all methods having very low ARI scores ranging from 0.058 to 0.153. For the data of the other three platforms, including ST, Stereo-seq, and SeqFish, the performance of different methods achieved a moderate level ([Supplementary Fig. S11](#)). For the three ST datasets, BayesSpace performed consistently better than other methods (0.642), followed by Leiden (0.562) and STMGCN (0.560). For Stereo-seq data, the performance of the top five methods (SpaGCN, Leiden, GraphST, Louvain, and BayesSpace) is similar, with an average ARI score ranging from 0.444 to 0.463. For SeqFish data, the three traditional methods (Louvain, Seurat, and Leiden) outperformed most GNN methods (mean ARI from 0.441 to 0.461), followed by STMGCN, BayesSpace, and SpaGCN (mean ARI from 0.382 to 0.440). Interestingly, Seurat, Louvain, and Leiden have poor overall rankings on 10X Visium data, while their rankings increase greatly and even outperform most GNN methods on SeqFish and Slide-seqV2 data. This phenomenon may be due to the high noise levels in SeqFish and Slide-seqV2 data, and GNN-based methods are inherently sensitive to noise, resulting in poor clustering performance on such data [59]. In contrast, traditional clustering methods are more robust to noise, thus performing relatively better on SeqFish and Slide-seqV2 data. Overall, these results suggest that no method can achieve the best performance on data from different SRT platforms, and there is still significant room for enhancing the performance of spatial domain recognition methods for STARmap and Slide-seqV2 data.

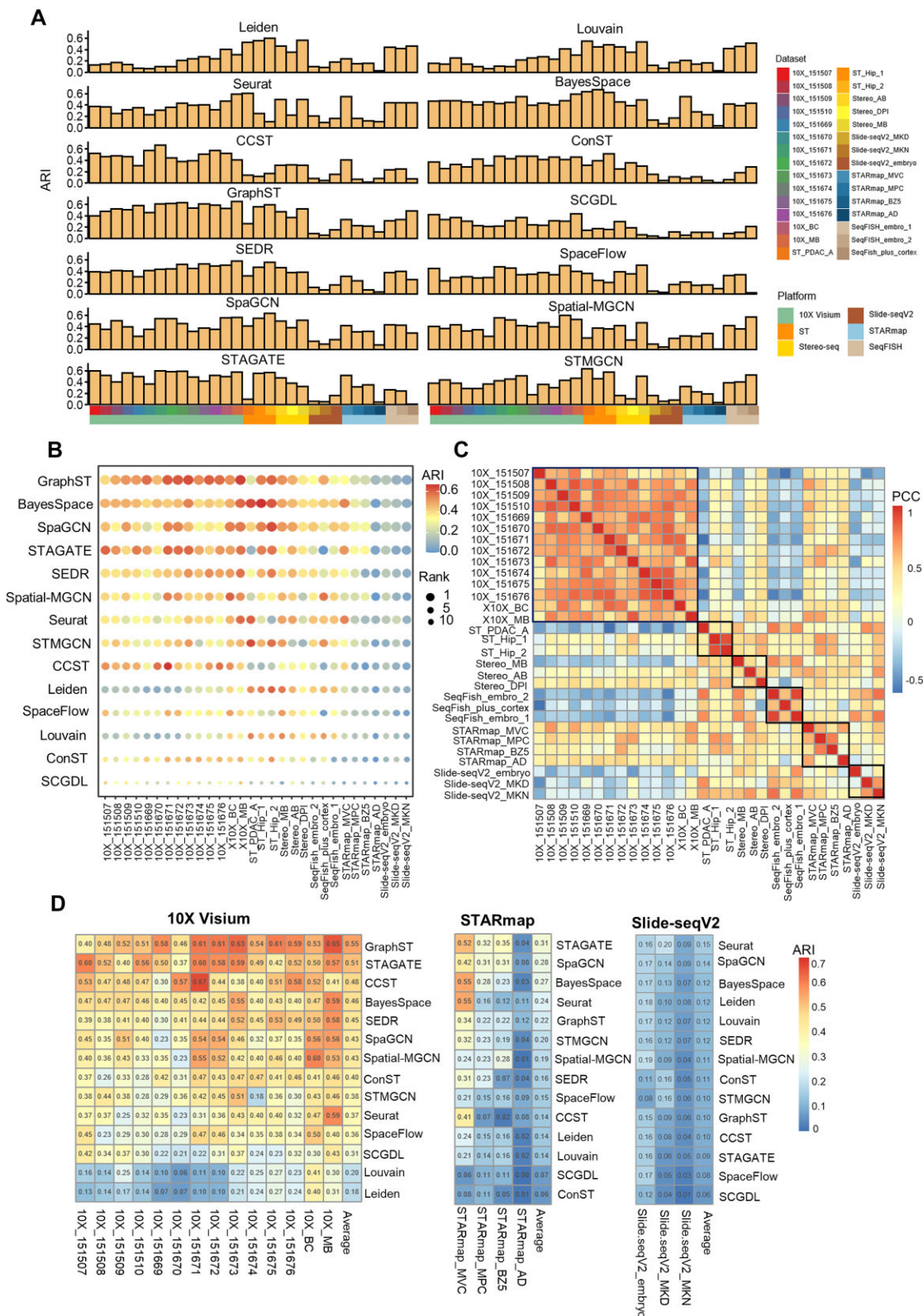


Figure 3. Evaluating the generalizability of methods in spatial domain identification using datasets from different SRT technologies. **(A)** ARI scores of different methods on different datasets. **(B)** Bubble chart showing ARI scores and rankings of different methods on different datasets. The ranking for each dataset is based on the ARI scores of different methods on the dataset. **(C)** Spearman correlation of ARI scores between datasets. For each pair of datasets, the correlation coefficient is calculated based on the ARI scores of all methods on the respective data. **(D)** Heatmaps showing the ARI scores and rankings of different methods on datasets from different SRT platforms including 10X Visium, STRAmap and Slide-seqV2. The display order of methods in each heatmap is based on the ranking of the average ARI score across all datasets.

Next, we assessed the impact of the choice of clustering method and refinement on spatial domain identification of 12 GNN methods capable of generating embedded representations. A total of eight configurations were generated, covering four clustering algorithms in default or refined mode (see “Materials and methods”). We found that the clustering algorithms used by default within the GNN methods are not always optimal (Supplementary Fig. S12). For example, DeepST performs worst with the default K-means clustering. It is also observed that the choice of clustering algorithm greatly impacts the performance of GraphST, DeepST, and SEDR, whereas it has a lesser effect on ConST and stLearn. In addition, when using the default clustering method, the refined mode can improve the clustering results in most cases. Among these GNN methods, after refinement, the performance of SpaGCN and STMCCN was improved the most, while the performance of ConST, stLearn, and CCST remained almost unchanged. Further, we evaluated the impact of three hyperparameters of GNN methods on spatial clustering, including the number of epochs, the number of low-dimensional embedding dimensions, and learning rate, using the DLPFC #151 673 dataset. Overall, different GNN methods have varying sensitivities to different parameters (Supplementary Fig. S13). For example, ARI scores of ConST and GraphST fluctuated much more with the change of number of epochs compared to other methods. Meanwhile, the ARI score of ConST also showed significant fluctuations with changes in the other two parameters. Among these methods, DeepST and Spatial-MGCN were relatively less affected by different parameters. Fortunately, the majority of methods obtained relatively high ARI scores under default parameter settings, indicating that the default parameter values of most methods have been optimized. Therefore, in general, users could directly use the default parameter settings when using these methods. However, due to the great impact of these parameters on most methods, especially the learning rate and the number of input dimensions, users can also adjust the parameters of the methods to meet specific needs.

Domain-specific SVGs detected based on spatial domains identified by different GNN methods have high accuracy but low concordance

With the guidance of the identified spatial domains, spatial domain-specific SVGs that exhibit enriched expression pattern in predefined spatial domains can be detected. Next, we evaluated the impact of spatial domains identified by different methods on domain-specific SVG detection. We selected the most representative datasets with superior clustering performance from each SRT platform for SVG evaluation, including 10X_151_673, ST_Hip_2, Stereo_AB, Slide-seqV2_embryo, STARmap_MVC, and SeqFish_embryo_2. Since real SVGs are unknown in most real data, the quantitative assessment of SVG identification on real data is challenging. We proposed a strategy to construct consensus SVGs as the reference (Supplementary Fig. S3), which were identified from three high-performance global SVG identification methods including SPARK-X [53], nnSVG [54], and Moran’s I [55] (see “Materials and methods”). We obtained 147 to 906 consensus global SVGs across the six real datasets (Fig. 4a). Moreover, to avoid bias in evaluation results due to missing some true SVGs of the consensus SVG set, we further proposed a refinement strategy to recruit identified SVGs that are not in the consensus set but have high Moran’s I score as TPs (see

“Materials and methods”). After refinement, we recruited 14 non-redundant SVGs identified by GNN methods across the six datasets (Supplementary Fig. S14). Upon these strategies and refinement, we believe that the high confidence consensus SVGs can serve as a reference for evaluating domain-specific SVGs. To further verify whether this is indeed the case, here we used the twelve DLPFC slices with 200 known SVGs, and obtained domain-specific SVGs as well as consensus SVGs. Results showed that although the number of consensus global SVGs is less than 200, they are all in the simulated reference SVGs, indicating that the consensus SVGs are true and conservative (Supplementary Fig. S15). More importantly, for domain-specific SVGs recognized by all GNN methods, as long as they are real (i.e. present in the 200 simulated SVGs), they are mostly in the consensus SVG set, indicating that the consensus SVGs always cover domain-specific SVGs. These results suggested the validity of using the consistent SVG set as a reference to evaluate domain-specific SVGs.

Having confirmed the effectiveness of consensus SVGs, next we evaluated domain-specific SVG recognition results. We followed the idea of SpaGCN, which has been adopted in various studies [38, 60, 61], for detecting domain-specific SVGs based on identified spatial domains from different GNN methods. Across all datasets, the numbers of SVGs identified based on spatial domains obtained from all method are significantly lower than the number of consensus SVGs (Fig. 4a versus Fig. 5a). Even using the union of SVGs detected by all methods, the number of detected SVGs is far less than the reference SVG set (Supplementary Fig. S16). Moreover, within the same dataset, the number of SVGs detected by different methods varies greatly (Fig. 4b). For instance, on the Stereo_AB dataset, Spatial-MGCN identified 336 SVGs, while CCST and ConST only obtained less than 10 SVGs. Interestingly, on the SeqFish_embryo_2 dataset, the number of SVGs identified by all methods was generally high, and all identified SVGs were in consensus SVGs. This is largely due to the fact that most of the genes in this dataset are all cellular markers that exhibit specific expression patterns in different cell types or regions [2]. Overall, only a very small number of SVGs detected by different methods are FPs (Fig. 4b). In addition, differential expression analysis of genes, similar to SVG analysis, is a routine step in many single-cell or spatial transcriptome analyses. By comparison, we also used the widely used tool Seurat to calculate differentially expressed genes (DEGs) between spots in a given domain and remaining spots in other domains. The number of DEGs is much higher than the number of SVGs (Fig. 4c), while the precision of the DEG results was significantly lower (Fig. 4d). This further illustrates the high specificity and precision of domain-specific SVG identification. Next, we used the agreement score [30] to measure the concordance of SVG identification by different methods on the same dataset. The consistency was relatively weak, with agreement scores ranging from 0.228 to 0.454 (Fig. 4e). Even for the SeqFish_embryo_2 dataset with known cellular markers, the agreement score is only moderate (0.454). This indicates great variability in SVGs identified by the various methods. Based on Moran’s I and Geary’s C scores that measure spatial autocorrelation of identified SVGs, the results obtained by the different methods were relatively similar, with CCST scoring slightly higher than the other methods (Supplementary Fig. S17). However, it should be noted that CCST typically detected fewer SVGs than the other methods, which introduces fewer FPs and results in the relatively high Moran’s I or Geary’s C score (Fig. 4b).

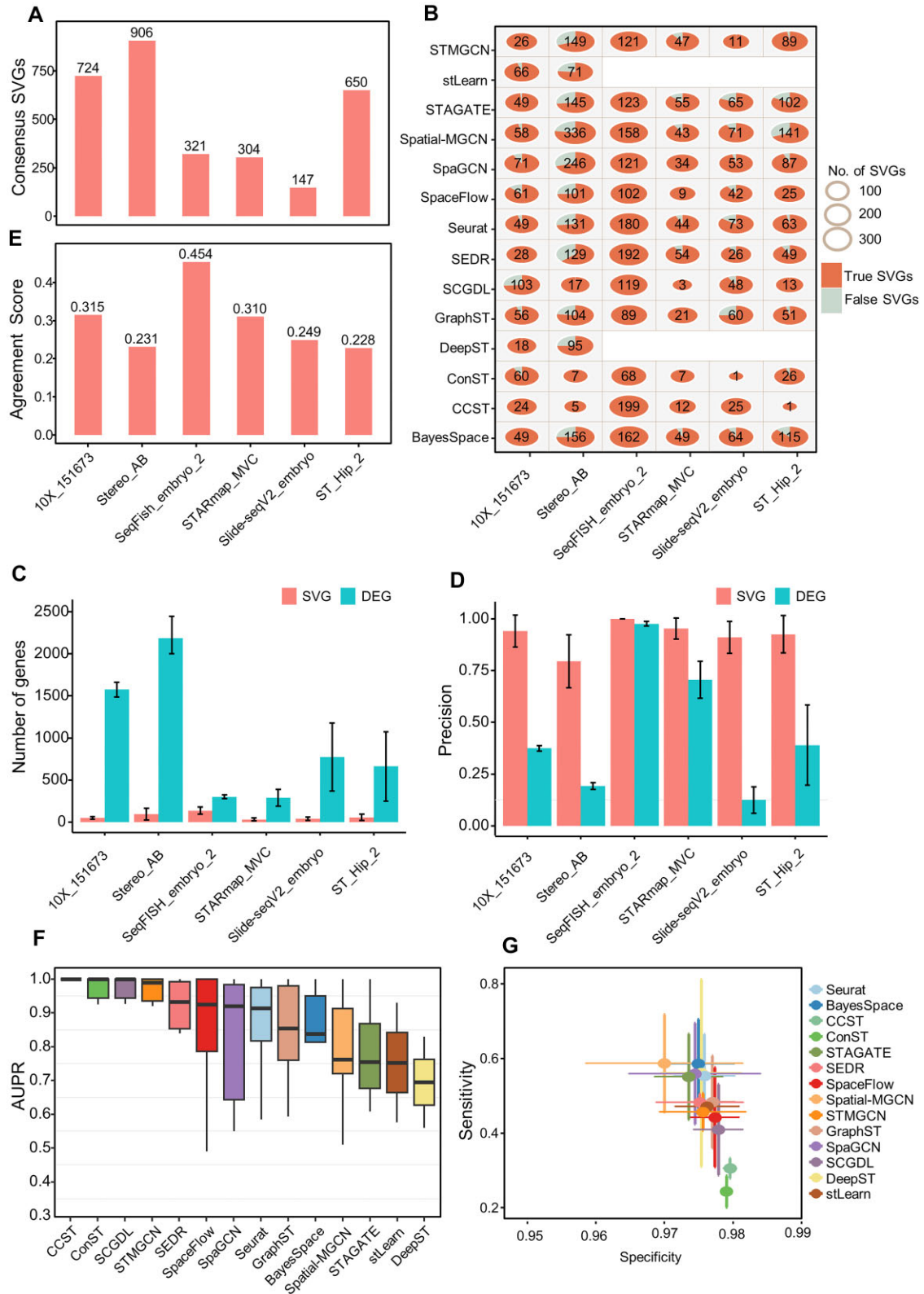


Figure 4. Evaluating spatial domain-specific SVG detection based on spatial domains identified by different methods using six representative real datasets. **(A)** Numbers of consensus SVGs. **(B)** Numbers of true and false SVGs detected based on spatial domains identified by different methods. **(C)** Numbers of SVGs and DEGs identified based on spatial domains identified by different methods. **(D)** Precision of SVG and DEG results. The consensus SVG list in **A** was used as the gold standard. **(E)** Agreement score that measures the concordance of SVG identification by different methods on the same dataset. **(F)** Boxplot summarizing AUPR scores of SVG results from each method on the six datasets. **(G)** Specificity and sensitivity of SVG results from different methods. The error bar displays the range of variation of each method's metric across the six datasets.

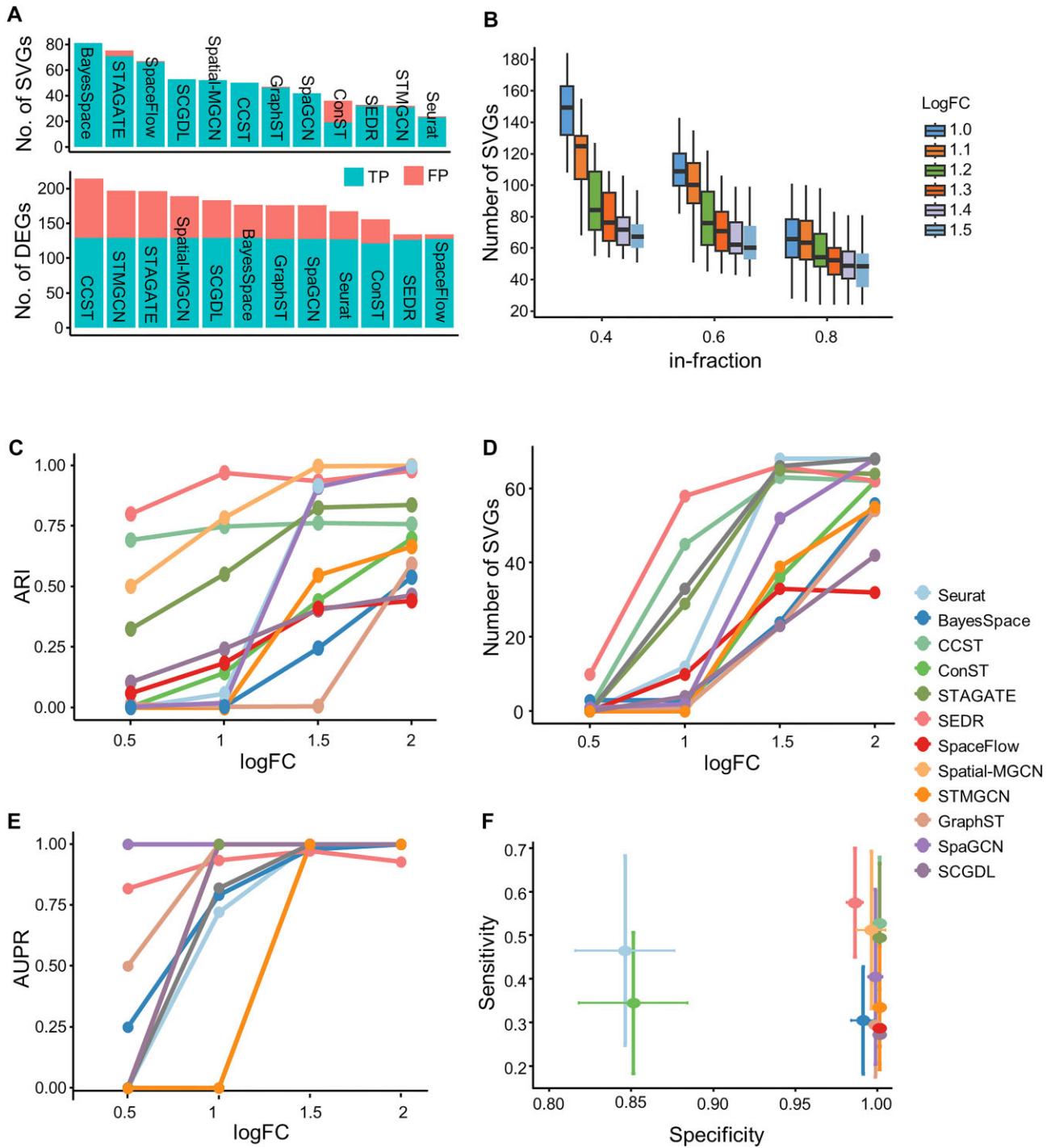


Figure 5. Evaluating spatial domain-specific SVG detection based on spatial domains identified by different methods using simulated datasets. **(A)** Numbers of true and false SVGs detected based on spatial domains identified by different methods on the DLPFC_scDesign3 simulated data. **(B)** Numbers of SVGs detected with different logFC and in-fraction thresholds on simulated DLPFC#151 673 slice. The boxplot summarizes the numbers of SVGs identified from different methods. **(C)** Variation of ARI scores for each method on the four Simu Dataset3 simulated with different logFC thresholds. **(D)** Same as c, except that number of SVGs was calculated. **(E)** Same as c, except that AUPR was calculated. **(F)** Specificity and sensitivity of SVG results from different methods. The error bar displays the range of variation of each method's metric across the four Simu Dataset3 datasets.

Therefore, using only metrics like Moran's I or Geary's C cannot comprehensively evaluate SVG results, such as SVG recognition accuracy or quantity.

Next, we comprehensively compared the accuracy of different methods in identifying SVGs using the AUPR metric. The overall AUPR scores of CCST, ConST, and STMGCN are higher than other methods (Fig. 4f). Particularly, AUPR score of CCST is 1, indicating that all SVGs identified by CCST are in the consensus SVG list. These methods show an overall good performance in specificity with an average value of 0.996 (Fig. 4g). This high specificity indicates that the likelihood of FPs in the SVGs identified by these methods is very low. However, their sensitivity is only moderate, with an average of 0.472, indicating that the used domain-specific SVG detection method, SpaGCN, can only capture a small number of true SVGs regardless of which GNN method is used to identify spatial domains. It is worth noting that no single method achieves a perfect balance between specificity and sensitivity. For instance, BayesSpace and Spatial-MGCN show the highest sensitivity but have relatively lower specificity, while ConST excels in specificity but has the lowest sensitivity (Fig. 4g).

Benchmarking spatial domain-specific SVG detection using synthetic spatial transcriptomics data

Next, we used several synthetic SRT datasets to further assess the impact of spatial domains identified by different methods on domain-specific SVG detection. First, we generated a simulated dataset based on the DLPFC data using scDesign3 [33], which includes 200 real SVGs and 1800 non-SVGs (data DLPFC_scDesign in [Supplementary Table S2](#)). The GNN methods only capture an average of 49 SVGs, less than 25% of the real SVGs (Fig. 5a). Consistent with the evaluation results using real data (Fig. 4c and d), only a small number of SVGs identified from the simulated dataset by different GNN methods except for ConST were FPs (Fig. 5a). In contrast, the number of DEGs identified by Seurat was more than three times that of domain-specific SVGs, and there were also more FPs (Fig. 5a). Interestingly, the number of SVGs recognized based on spatial domains identified by different GNN tools varies greatly, while the number of DEGs, especially TPs, is relatively close. This indicates that the SVG detection method of SpaGCN is more conservative than the DEG detection method of Seurat, and also more susceptible to the given spatial domains. Next, we further conducted experiments on simulated datasets with varying numbers of reference SVGs ranging from 100 to 900 (data DLPFC_scDesign_2). Similarly, the number of DEGs identified is much higher than the number of SVGs on the same simulated data, but there are also more FP DEGs ([Supplementary Fig. S18](#)). Moreover, as the number of reference SVGs increases, the number of identified SVGs and DEGs increases, saturating on the dataset with 700 reference SVGs. This may be due to the fact that the more reference SVGs a dataset has, the more accurate the spatial domain obtained by GNN tools may be, resulting in relatively more SVGs or DEGs. Summarizing all domain-specific SVGs or DEGs identified by all GNN tools on each simulated dataset, the average recall using the combined domain-specific SVGs is 33.6%, which is much lower than the 60.9% of DEGs ([Supplementary Fig. S19](#)). However, the average precision using the combined domain-specific SVGs is higher (SVG = 82%

versus DEG = 69.7%). Overall, these results suggested that domain-specific SVGs detection method of SpaGCN is sensitive to given spatial domains and has high precision but low sensitivity.

Further, we examined the impact of two parameters of the SVG detection method in SpaGCN [46], logFC and in-fraction. The logFC parameter is the threshold of the log expression fold change between the target and neighboring domain. The in-fraction parameter is the threshold for the percentage of spots expressing the gene in the target domain. Here the logFC parameter was evaluated over an increasing range from 1.0 to 1.5, with increments of 0.1, resulting in six different scenarios. The in-fraction parameter increased from 0.4 to 0.8 in steps of 0.2, covering three scenarios. Both parameters have a significant effect on the number of identified SVGs (Fig. 5b). Additionally, as the in-fraction and logFC increase, the numbers of TPs and FPs, as well as the TP rate, generally decline, while the AUPR consistently rises ([Supplementary Fig. S20](#)). Overall, as the values of in-fraction and logFC increase, the accuracy of SVG detection increases, but the number of identified SVGs decreases significantly. Therefore, in practical applications, these parameters need to be adjusted carefully to find a suitable balance between the accuracy and the number of SVG recognitions.

To further examine the impact of the logFC on the number of identified SVGs, we used an additional synthetic dataset that simulating gene expression data with four different logFC levels [10] (data BC_SRTsim). First, we identified spatial domains using different spatial clustering methods. With the increase of logFC, ARI scores were improved for most methods, except for CCST and SEDR (Fig. 5c). This suggests that increasing the level of gene expression variation helps to improve the spatial domain identification for most methods. Meanwhile, the number of SVGs detected by most methods also increases significantly with the increase of logFC (Fig. 5d). Specifically, some methods, such as SEDR and CCST, Spatial-MGCN, and STAGATE, have relatively high ARI scores when logFC is low (0.5), while almost all methods obtain a very small number of SVGs (<10) at this threshold. According to the AUPR score, SpaGCN, SEDR, and GraphST have relatively good performance, especially when logFC is small (Fig. 5e). The AUPR score for each method steadily increased as the increase of logFC. It is worth noting that when LogFC is 0.5, some methods such as SpaGCN and SEDR have higher AUPR scores (Fig. 5e), but this may be due to the extremely small number of SVGs recognized (Fig. 5d). In contrast, methods such as GraphST and STAGATE require LogFC to reach 1.0 to achieve high AUPR scores. These results suggest that different methods require some degree of gene expression variation to effectively recognize SVGs. Overall, specificity is high for most methods (>0.95), except for ConST and Seurat (<0.9), which implies a relatively larger number of FP SVGs detected based on the spatial domains identified by ConST and Seurat. In terms of sensitivity, all methods are not high, among which SEDR, CCST, and Spatial-MGCN have relatively higher sensitivity, indicating their relatively strong ability to detect true SVGs (Fig. 5f).

Spatial domain identification results affect spatial domain-specific SVG detection

Spatial domain-specific SVGs are detected based on predefined spatial domains, next we examined the impact of spa-

tial domain recognition results on the detection SVGs. Our above evaluation has revealed that the domain-specific SVGs detected based on spatial domains identified from different methods are generally true (Figs 4d and 5a). However, it is still unknown how the spatial domain recognition results affect the number of detected SVGs. To answer this question, we adopted three simulated data (containing 14 datasets) and six real data for subsequent evaluation. Spatial domains were identified by each of the 12 spatial clustering methods, and then domain-specific SVGs were detected based on the identified spatial domains. Of note, GNN methods for spatial clustering often involve gene selection before identifying spatial domains; some methods use all genes as input for clustering, such as ConST, SEDR, and SpaGCN, while some other methods use HVGs, like GraphST, STAGATE, and STMGCN (see Supplementary Table S4). First, we examined the potential impact of the upstream gene selection method on downstream clustering and SVG identification. We conducted 12 GNN methods for spatial clustering across six real representative datasets with two different inputs: all genes and top 3000 HVGs. Results showed that in most cases the clustering results using the default input of the respective GNN method are slightly better than those using the other input (Supplementary Fig. S21a). The impact of input on different methods varies slightly; methods such as ConST, Spatial-MGCN, and BayesSpace are slightly more affected by the input than other methods. Moreover, in most cases, better clustering results correspond to slightly more recognized domain-specific SVGs (Supplementary Fig. S21b). Next, we further examined whether HVGs used for clustering tend to be recognized as SVGs. The results showed that regardless of using all genes or HVG as inputs, there was no significant difference in the proportion of identified SVGs that belonged to HVGs, indicating that the feature genes used for clustering were not associated with the identified SVGs (Supplementary Fig. S21c).

Next, we further explored the relation of the accuracy of the spatial domain identification results with the number of true SVGs identified using both real and simulated datasets. The results on all datasets showed a significant correlation between ARI scores and the number of real SVGs (Spearman $\rho = 0.502$, P -value $< 2e-16$) (Fig. 6a). Further, we divided the ARI scores obtained by the 12 methods on each dataset into four groups (three methods per group), and then compared the number of true SVGs detected by the groups with the highest and lowest ARI. It was found that in all datasets, except for 10X_151_673, the number of SVGs corresponding to the group with higher ARI was significantly higher than that of the group with lower ARI (Wilcoxon test P -value = 0.0005). (Fig. 6b). Similarly, we utilized the SC metric to quantify the spatial domain identification results and analyzed its relationship with the number of true SVGs. The impact of SC score on number of SVGs is not as significant as that of ARI score. The Spearman's correlation coefficient between SC and real SVG number was 0.208 (P -value = 0.001) (Supplementary Fig. S22a). Although the group with higher SC scores generally identified a higher number of SVGs compared to that with lower SC scores (Wilcoxon test P -value = 0.02), this difference was not as statistically significant as the effect of ARI on the number of real SVGs (Supplementary Fig. S22b).

Next, we further evaluated the impact of the spatial domains identified by different methods on downstream SVG identification. Overall, if a method identifies spatial domains more accurately (with higher ARI), then relatively more SVGs

can be identified based on its spatial domain results (Fig. 6c). For instance, ConST had the lowest average ARI (0.246), and the lowest number of SVGs were identified based on its spatial domains. In contrast, SEDR had the highest average ARI (0.50), corresponding to the highest number of SVGs. Moreover, there are also some methods, such as Seurat and BayesSpace, that only obtain moderate ARI values (~ 0.4), but their results contribute to detecting relatively high number of SVGs. However, it is worth noting that the effectiveness of different methods in spatial domain recognition on different datasets and their impact on the number of SVGs detected downstream vary (Supplementary Figs S23–S24). For instance, for simulated datasets based on the same DLPFC data but simulated with different tools, DLPFC_SRTsim and DLPFC_scDesign, GraphST obtains the highest ARI on the DLPFC_scDesign data with only a moderate number of SVGs (Supplementary Fig. S23a), whereas GraphST obtains low ARI on the DLPFC_SRTsim data with the lowest number of SVGs (Supplementary Fig. S23b). For another simulated dataset, BC_SRTsim, the trend in the number of SVGs with ARI was much more pronounced (Supplementary Fig. S23c). In comparison, for real datasets, although the number of identified SVGs generally increased with ARI, the trend was less pronounced (Supplementary Fig. S23). Specifically, for the real dataset 10X_151_673, there is no evident correlation between ARI and the number of SVGs (Supplementary Fig. S24). For instance, SCGDL has a relatively low ARI, but the spatial domains identified contribute to detecting the highest number of SVGs. In contrast, methods such as GraphST, Spatial-MGCN, and ConST exhibit considerable variation in ARI scores, while the corresponding SVG numbers are relatively similar. In addition, we conducted a similar analysis across all datasets to examine the impact of ARI on the accuracy of downstream SVG detection. Overall, the accuracy for SVGs identified based on spatial domains obtained by all methods was high (> 0.75) (Fig. 6d). Among these methods, CCST identified a moderate number of true SVGs (Fig. 6c), but it had the highest average accuracy of SVG detection (0.94) and a relatively high ARI score (0.487) (Fig. 6d). Methods like SEDR, which identified the highest number of true SVGs, also achieved high accuracy (> 0.85). In contrast, STAGATE identified a comparable number of SVGs to SpaGCN or SEDR (Fig. 6c), but had a lower accuracy of SVG detection (0.79) (Fig. 6d). These results indicate that overall, spatial domain recognition results with higher ARI scores help identify more SVGs with higher accuracy, but the impact of spatial domain on SVG recognition still depends on specific data and methods used.

Due to the significant differences in clustering performance of different GNN tools on different data, which further affects downstream SVG recognition, we attempted to ensemble spatial domain results identified by different GNN tools to see if we could obtain more robust clustering and SVG results. We employed a voting method from ensemble learning to integrate clustering results from multiple GNN methods. Since clustering labels obtained from different GNN methods are only numbers and do not directly correspond to real-world category labels, we employed a majority voting approach to map the predictions from each method onto the same space of true labels. After aligning labels from different methods, we selected the most frequently occurring label as the final category label for each spot and then calculated the ARI score. We performed the above ensemble clustering on six real datasets. It can be seen that the GNN methods with

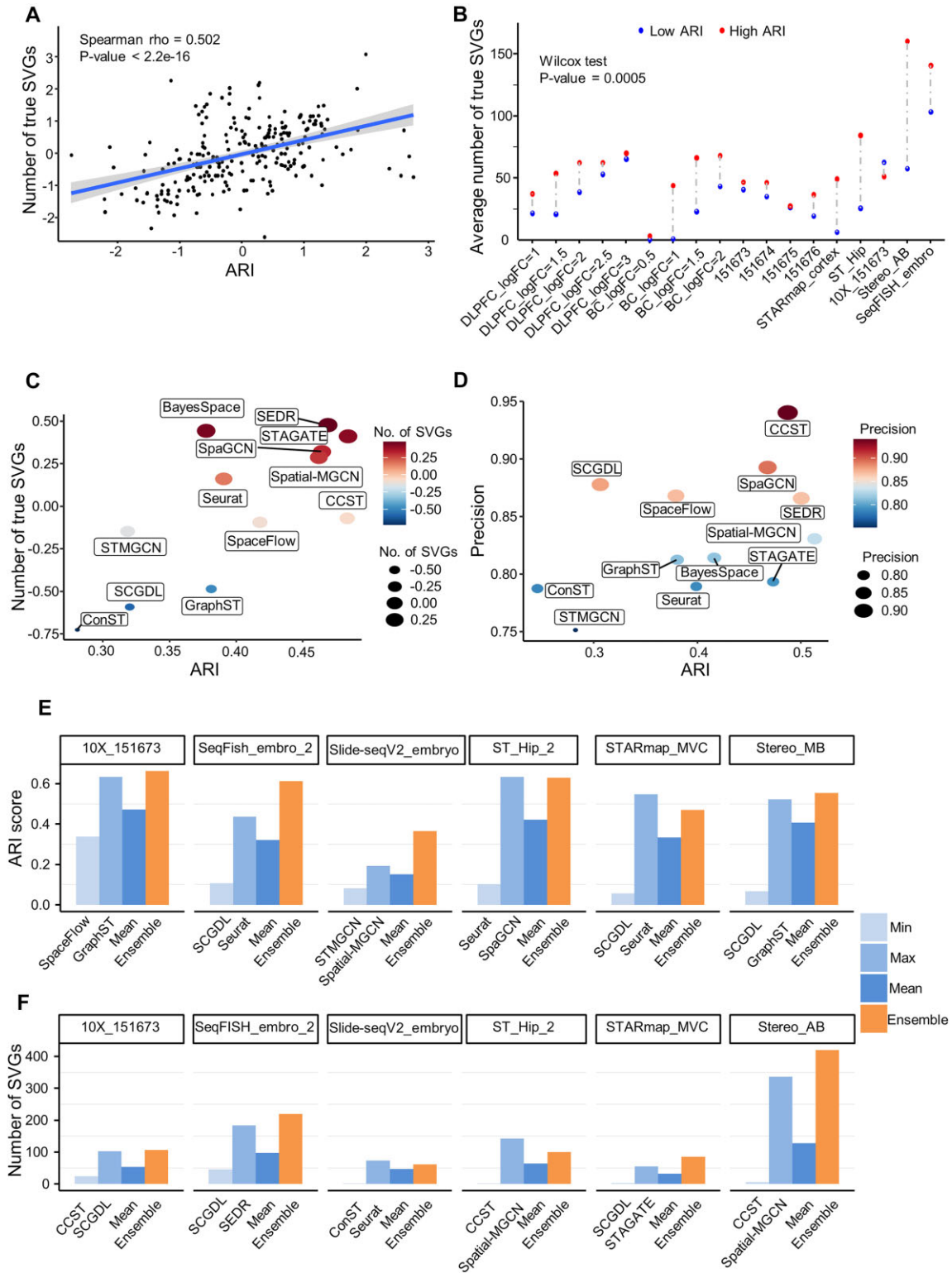


Figure 6. Evaluating the impact of spatial domains on spatial domain-specific SVG detection. **(A)** Correlation between ARI scores of spatial domains and number of detected true SVGs. A total of 6 real datasets and 14 simulated datasets were used. Each dot represents the average results of 12 GNN methods on the respective dataset. In order to make different data comparable, the number of SVGs and ARI scores were standardized. **(B)** Comparison of the number of true SVGs identified from spatial domains with high ARI and low ARI. For each dataset, 12 GNN methods were used for identifying spatial domains and ARI scores were calculated. The results for each dataset were categorized into four groups according to the ARI scores, and the group with the highest ARI and lowest ARI were retained for further analysis. Based on spatial domains in each group, domain-specific SVGs were obtained and the number of true SVGs was recorded. **(C)** Scatter plot showing the average ARI score and standardized number of true SVGs of each method across all datasets. The ARI score was calculated based on spatial domains identified from each method. **(D)** Scatter plot showing the average ARI score and precision of SVGs of each method across all datasets. **(E)** ARI scores of integrating clustering results from 12 GNN methods across six real datasets. Minimum, maximum, and average ARI scores of the 12 GNN methods were also provided. The method names with the minimum and maximum ARI scores are given on the X-axis. **(F)** Same as in **(E)**, except that numbers of SVGs were calculated.

the best (or worst) clustering or SVG recognition performance are not consistent different datasets (Fig. 6e and f), once again indicating that no method is optimal for all data. Moreover, clustering results with higher ARI tend to yield more downstream SVGs, which is consistent with above analysis results (Fig. 6a). Specifically, on all data, results obtained through ensemble clustering were better than the average results of all tools. In 4 out of 6 datasets, the integrated results were better than those of the optimal GNN method. Even for the two datasets where the integrated ARI scores are not the best (ST_Hip_2 and STARmap_MVC), the integrated ARI scores are still very close to the optimal ARI scores. These results indicate that integrating results from different GNN methods can lead to more robust and generally better clustering and SVG results, providing a new perspective for the development of future methods.

Computational time and memory usage

To evaluate the computational efficiency of methods on datasets from different SRT platforms, we conducted clustering experiments on six representative SRT datasets, including 10X_151_673, ST_Hip_2, Stereo_AB, Slide-seqV2_embryo, STARmap_MVC, and SeqFish_embryo_2. This evaluation was conducted on an Ubuntu operating system platform with hardware configurations that include an Intel(R) Xeon(R) Gold 6248R (3.00 GHz, 24-core) CPU, 256GB of memory, and an NVIDIA GeForce RTX 3060 GPU. In terms of time consumption, CCST took the longest time as it ran for 5000 epochs (Fig. 7a). Whereas, methods such as STMGCN, GraphST, and SEDR show less time consumption. In addition, SeqFish_embryo dataset requires longer processing times for most methods. This is attributed to the fact that this dataset has the highest number of spots, leading to a significant increase in computation when constructing graph neighbor relationships. In terms of memory consumption, STAGATE and DeepST have higher average consumption than other methods (Fig. 7b). In contrast, two traditional methods, Seurat and BayesSpace, show lower memory consumption.

Discussion

In this study, we conducted an in-depth evaluation of 19 methods for spatial domain identification from four key aspects, including accuracy, stability, generalizability, and scalability. Our results reveal that there is no single method that works best for all datasets, and the optimal method depends on the data (Fig. 3). Overall, most methods generally have high performance on 10X Visium data, while they generally have the much lower performance on STARmap and Slide-seqV2 data (Fig. 3d). Similar observations were found in previous studies [31, 62]. This may be because 10X Visium is currently the most widely used SRT platform, and most methods prioritize algorithm design for commonly used datasets, which may not generally effective for less-studied datasets [62]. This may also be due to the presence of continuous spatial domains and clearer boundaries in the benchmarked 10X Visium data, making it easier for clustering methods to achieve good results [31]. Considering all SRT platforms investigated, GraphST, BayesSpace, SpaGCN, and STAGATE are the most effective and show good robustness across various platforms in identifying spatial domains (Fig. 3b). Among them, BayesSpace has longer processing times and STAGATE has higher memory us-

age (Fig. 7), which may not be suitable for data with large scale. GraphST not only excels in spatial domain identification but also achieves the best balance between computational time and memory consumption, making it the preferred choice, followed by SpaGCN. Based on our platform-specific evaluation results, platform-specific method recommendations can also be provided (Fig. 3d). However, for STARmap and Slide-seqV2 data, the performance of all methods is generally poor. Nevertheless, we found that in most cases, integrating the results of multiple methods can be very beneficial (Fig. 6e). We also found that the default settings within the GNN methods are not always optimal (Supplementary Fig. S12). Therefore, we encourage users with certain programming skills to conduct more testing on specific data beyond the default settings of the methods to determine the optimal environment for each method. We also recommend that users run multiple methods and integrate the results of different tools to achieve more robust clustering results, particularly for non 10X Visium data. We have provided relevant scripts on our GitHub site for users to refer to. A practical guide for users to select appropriate methods for diverse scenarios is provided in Fig. 7c.

Although current GNN methods have made good progress in spatial domain recognition tasks, there is still significant room for improvement in performance on diverse datasets and many challenges remain. First, SRT data are complex and characterized by high dimensionality, high sparsity, and noise interference, which may mask the spatial expression patterns of genes and limit the performance improvement of clustering techniques. Second, current methods lack effective strategy to extract and integrate features from H&E images. In clustering experiments on DLPFC dataset, we found including spatial location information can significantly improve the results, but introducing H&E image information may not necessarily improve the results (Supplementary Fig. S5e). Current methods for extracting information from H&E images involve segmenting the image by spots/cells and capturing features through various feature extraction modules, such as ResNET [50] and Vision Transformers [51, 52]. These methods show potential in extracting H&E image features, but how to accurately segment the image and accurately align the segmented spots/cells with SRT data remain technical challenging [63]. In light of the aforementioned issues, exploring advanced techniques for processing histological images is vital. It is also important to capture complementary information from additional modalities such as ATAC (Assay for Transposase Accessible Chromatin), proteomics, and lipidomics to establish a more comprehensive understanding of spatial genomics. In addition, integrating the results of different GNN tools through more effective ensemble learning to improve robustness and accuracy of spatial clustering could be also a promising research direction.

In this study, we also investigated the impact of identified spatial domains from different spatial clustering methods on domain-specific SVG identification. Because there is no reference standard of SVGs in most real data, the quantitative assessment of SVG identification results on real data is challenging. To address this challenge, we first constructed the consensus SVG set using multiple global SVG methods as the gold standard, and further employed a refinement strategy to compensate for the incompleteness of the consensus SVG set. We comprehensively evaluated the rationality and effectiveness of the constructed consensus SVGs for evaluating domain specific SVGs (Supplementary Figs S1–S4, S14–S15), which

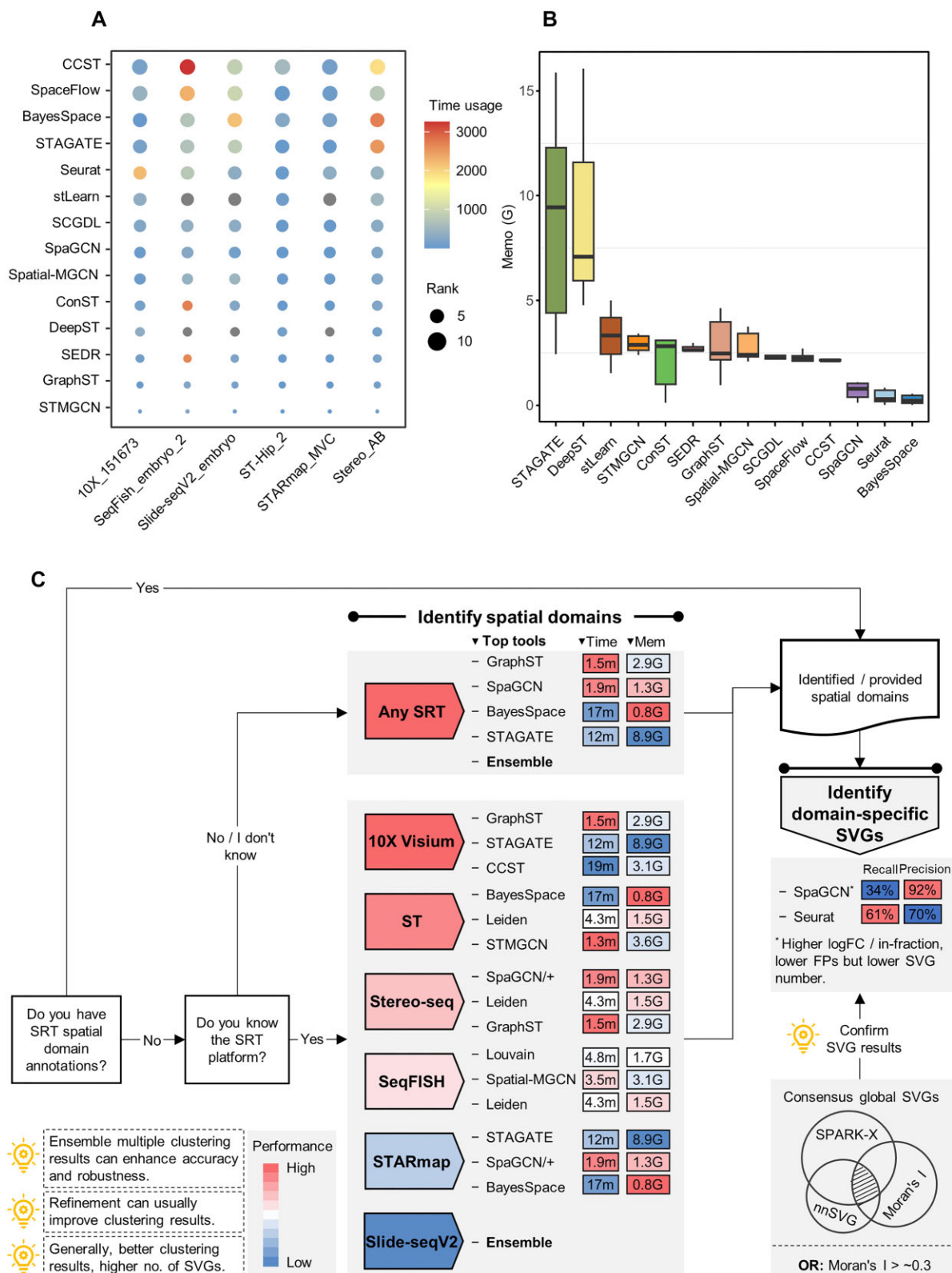


Figure 7. Comparison of time and memory consumption of different methods. **(A)** Time consumption of benchmark methods on six representative datasets. Rankings of methods were sorted in descending order based on the average time consumption across datasets. **(B)** Boxplot summarizing the memory consumption of different methods across the six datasets. **(C)** A practical guide for users to select appropriate methods for diverse scenarios. Time and memory are the average of the corresponding method on six real datasets.

also provided important references for the performance evaluation of other domain-specific SVG methods in the future. Our benchmark results reveal that identifying SVGs based on the SpaGCN pipeline is characterized by high specificity but low sensitivity (Figs 4g and 5f). Almost all domain-specific SVGs obtained from spatial domains identified by different methods are true SVGs (Fig. 5a). However, domain-specific SVG identification is highly conservative, capturing far fewer SVGs than global SVG or DEG methods (Fig. 4c and d and Supplementary Figs S18–S19). The SpaGCN pipeline employs strict threshold criteria when identifying SVGs, while reducing the FP rate, also sacrifices some sensitivity. This trade-off is generally acceptable, especially in situations where highly reliable results are required. It can significantly reduce the number of candidate genes, thereby highlighting those most likely to be prioritized for further investigation. Nevertheless, it is also necessary to propose more sensitive methods that can identify more domain-specific SVGs while ensuring accuracy. Additionally, we need to point out that users could choose different types of SVGs (e.g. domain-specific or global SVGs) based on their research objectives. Domain-specific SVGs are genes that exhibit significant differential expression within predefined spatial domains. This allows the identified SVGs can be associated with the biological significance of spatial regions. In contrast, global SVGs are typically identified through statistical models, which are recognized for their variation across the entire tissue, not confined to specific regions. Due to more stringent criteria, the number of domain-specific SVGs obtained is significantly lower than global SVGs. In addition, in this study, we also used Seurat to calculate the DEGs between spots in one domain and spots in remaining domains, which can in principle be considered as domain-specific SVGs. Through evaluation, we found that the Seurat method has higher sensitivity but lower specificity compared to the SpaGCN method (Supplementary Figs S18–S19). As a practical guide, if users want to recognize as more SVGs as possible, they can consider using global SVG recognition methods. If users are interested in genes specific to given tissue domains and want to reduce the risk of FPs, then domain-specific SVG recognition methods would be more suitable. As a supplement, users can also use Seurat to obtain more domain-specific SVGs, but attention should be paid to potential false predictions.

Currently, a few methods are available for domain-specific SVG detection [54]. We used the function provided in SpaGCN for domain-specific SVG recognition, which has been adopted in various studies [38, 60, 61]. Of note, we did not compare different domain-specific SVG detection methods, but aimed to assess the impact of spatial domains identified from different GNN methods on the downstream SVG recognition. We found that to some extent, better spatial domain recognition results (higher ARI and/or SC scores) contribute to detecting more SVGs with higher accuracy (Fig. 6). However, the impact of spatial domain on SVG recognition is still related to the data to be analyzed and the methods used. There are still some methods that can achieve good spatial domain detection results on certain datasets, but the resulting SVG is rare (Supplementary Figs S23–S24). These evaluation results indicate that although the recognition performance of SVGs is related to spatial clustering results, there are still other influencing factors. We also investigated the two important parameters of SpaGCN's SVG recognition function used in this study, logFC and in-fraction, and found that both parameters have a significant impact on the number and accuracy of SVGs

(Fig. 5b–e). Especially, the differences in gene expression levels between different domains have a direct impact on spatial clustering and SVG recognition (Fig. 5c–e). Therefore, in practical use, these parameters should be carefully set.

There are some limitations of this study. Emerging state-of-the-art methods such as stCluster [64], SpaNCMG [65], and IRIS [63] were not included in the preparation. Additionally, we only considered the SpaGCN method for SVG detection in this work, and more different spatial domain-specific SVG recognition methods could be considered in the future to compare how clustering results affect SVG recognition under different spatial domain recognition frameworks. Moreover, this study focused on the evaluation of spatial domains or embeddings obtained from different method on two fundamental tasks: spatial domain identification and spatial-domain specific SVG detection. In further work, additional downstream tasks, such as trajectory inference, batch correction, and cell-cell communication, could also be considered.

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Supplementary data

Supplementary data are available at NAR online.

Conflict of interest

None declared.

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Data availability

All the data used in this study are publicly available. All scripts involved in this study can be downloaded from https://github.com/BMILAB/Benchmark_ST_analysis and <https://zenodo.org/records/15114362>.

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